

Learning Parity with Quantization: Achieving Full-Rate Encryption by Exploiting Quantization Noise in Code-Based Cryptography

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Abstract. The Learning Parity with Noise (LPN) problem has become a cornerstone for building lightweight, post-quantum secure encryption schemes. Despite its widespread adoption, LPN-based constructions suffer from a fundamental efficiency limitation: the essential noise term that provides security simultaneously requires error correction coding, leading to bandwidth overhead. We introduce a variant of LPN termed Learning Parity with Quantization (LPQ). While maintaining the “learning from noisy equations” framework, LPQ generates Bernoulli-like noise from code-aided quantization and enables simultaneous security and compression. Formally, the $\text{LPQ}_{N,n,\mathcal{C}}$ problem challenges adversaries to distinguish the triplet $(\mathbf{A}, Q_{\mathcal{C}}(\mathbf{A}\mathbf{s} \oplus \mathbf{u}), \mathbf{u})$ from uniform, where $Q_{\mathcal{C}}$ is a vector quantization function based on an (N, K) code $\mathcal{C}$, and $\mathbf{u}$ serves as a public dither. We establish the hardness of LPQ through a tight reduction from the LPN problem, maintaining equivalent security guarantees. We demonstrate LPQ’s practical efficacy through a full rate (i.e., rate-1) symmetric key encryption scheme, where LPQ combined with an extendable output function (XOF) achieves optimal ciphertext efficiency ($|\text{ct}| = |\text{pt}|$).

Keywords: Learning Parity with Noise (LPN) · Polar Code · Quantization · Ciphertext Compression

1 Introduction

The Learning Parity with Noise (LPN) problem constitutes a fundamental building block for code-based cryptography, providing lightweight security rooted in the hardness of decoding random linear codes. Formally, given a security parameter n and noise rate $\mu \in (0, 1/2)$, the LPN problem requires recovering a secret $\mathbf{s} \in \mathbb{F}_2^n$ from samples of the form:

$$(\mathbf{A}, \mathbf{b} = \mathbf{A}\mathbf{s} \oplus \mathbf{e}) \in \mathbb{F}_2^{N \times n} \times \mathbb{F}_2^N \quad (1)$$

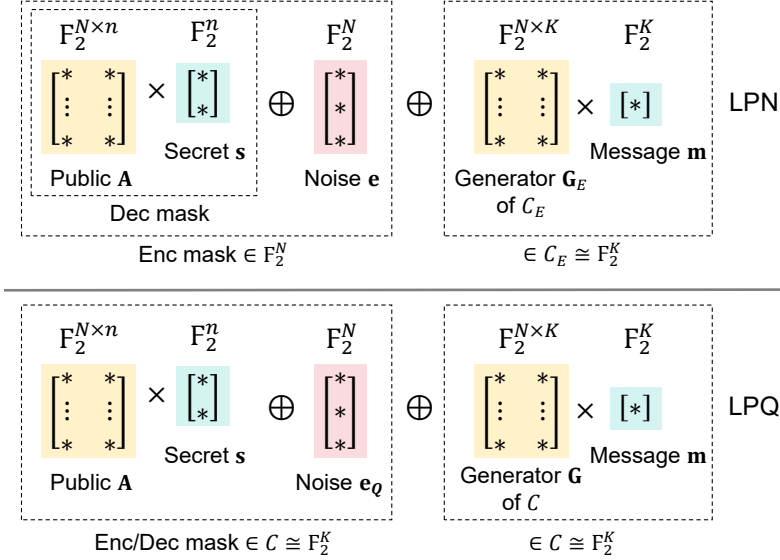


Fig. 1. Comparison of LPN (error correction via code $\mathcal{C}_E$) and LPQ (quantization via code $\mathcal{C}$) encryption schemes.

where $\mathbf{A} \xleftarrow{\$} \mathbb{F}_2^{N \times n}$, $\mathbf{e} \sim \text{Bern}(\mu)^N$, and $N = \text{poly}(n)$. Its decisional variant (DLPN)—distinguishing these samples from uniform random pairs—forms the security foundation for numerous cryptographic constructions.

Despite their provable security [BKL21], LPN-based encryption schemes suffer from inherent efficiency constraints: Security-critical noise vector $\mathbf{e}$ necessitates error correction coding; in terms of symmetric encryption, the code rate $R = |\text{pt}|/|\text{ct}|$ is fundamentally limited by $1 - H_2(\mu)$, corresponding to the capacity of a *binary symmetric channel (BSC)*. This creates a fundamental tension: increasing noise ($\mu \uparrow$) enhances security at the expense of rate, while decreasing noise ($\mu \downarrow$) improves bandwidth efficiency but weakens security guarantees.

In this work, we introduce the Learning Parity with Quantization (LPQ) problem, a new paradigm of learning from noisy linear equations where the noise is from lossy compression. As illustrated in Fig. 1, the key differences between LPN and LPQ are:

$$\begin{aligned}
 & \text{(LPN Mask)} \quad \mathbf{b} = \mathbf{A}\mathbf{s} \oplus \mathbf{e} \in \mathbb{F}_2^N \\
 & \text{(LPN Encryption)} \quad \underbrace{\mathbf{A}\mathbf{s} \oplus \mathbf{e}}_{\text{encryption mask} \in \mathbb{F}_2^N} \oplus \underbrace{\mathbf{G}_E \mathbf{m}}_{\text{encoded message} \in \mathcal{C}_E} \\
 & \text{(LPQ Mask)} \quad \mathbf{b} = \mathbf{A}\mathbf{s} \oplus \mathbf{e}_Q \in \mathcal{C} \\
 & \text{(LPQ Encryption)} \quad \underbrace{\mathbf{A}\mathbf{s} \oplus \mathbf{e}_Q}_{\text{encryption mask} \in \mathcal{C}} \oplus \underbrace{\mathbf{G} \mathbf{m}}_{\text{encoded message} \in \mathcal{C}}
 \end{aligned}$$

While both schemes employ an (N, K) linear code, their purposes differ fundamentally: in LPN, $\mathcal{C}_E$ corrects transmission errors, whereas in LPQ, $\mathcal{C}$ enables ciphertext compression through vector quantization.

The mask structures reveal fundamental operational differences between the schemes: In LPN, the encryption mask $\mathbf{A}\mathbf{s} \oplus \mathbf{e}$ differs from the decryption mask $\mathbf{A}\mathbf{s}$, requiring error correction for message recovery, while LPQ uses identical masks $\mathbf{b} = Q_{\mathcal{C}}(\mathbf{A}\mathbf{s} \oplus \mathbf{u})$ for both encryption and decryption, where $\mathbf{u}$ is a public dither and $Q_{\mathcal{C}}$ implements $\mathcal{C}$ -based vector quantization. This structural symmetry eliminates error-correction needs.

The information rate, defined as the ratio of plaintext size to ciphertext size, demonstrates LPQ’s efficiency: LPN achieves rate $\frac{K}{N}$ due to error-correction overhead, whereas LPQ attains the optimal rate 1 by maintaining isometry between plaintext and ciphertext spaces.

1.1 Our Contributions

By properly designing the quantizer $Q_{\mathcal{C}}$ to enable $\mathbf{e}_Q$ featuring the Bernoulli noise of LPN, the proposed LPQ problem enables compact representations while preserving cryptographic hardness of LPN.

Our contributions are threefold:

1. **Code-Based Dithered Quantization.** We propose a quantization framework that replaces Bernoulli noise $\mathbf{e}$ with quantization noise $\mathbf{e}_Q$ derived from binary linear codes. By applying a random dither prior to quantization, we ensure the quantization noise is statistically independent of the input, a critical property for security in cryptographic settings. When instantiated with rate-distortion bound achieving polar codes, this construction achieves computational indistinguishability from Bernoulli noise while supporting $O(N \log N)$ -time encoding/decoding.

2. **Tight Security Reduction from LPN to LPQ.** We establish a tight security reduction proving that any distinguisher for LPQ samples can be transformed into an adversary against the classical LPN problem. This reduction preserves the problem’s cryptographic hardness, demonstrating that structured quantization noise inherits the security guarantees of Bernoulli noise. Our result bridges code-based quantization with LPN’s well-studied security assumptions, offering new insights into leveraging coding-theoretic structure for cryptographic design.

3. **Bandwidth-Optimal Encryption from LPQ.** We construct a private-key encryption scheme where ciphertexts achieve minimal overhead: $|\text{ct}| = |\text{seed}| + |\text{pt}|$. By generating the public matrix $\mathbf{A}$ and dither vector $\mathbf{u}$ from a single seed via an extendable output function (XOF, e.g., SHA-3), our scheme eliminates redundant storage and ensures the encryption rate $R = \frac{|\text{pt}|}{|\text{ct}|}$ asymptotically approaches 1. This optimal rate – unachievable in standard LPN-based encryption – reduces communication overhead while maintaining provable security under LPQ assumptions.

1.2 Technical Overview

Our framework integrates two techniques: code-based dithered quantization for lossy compression and polar code as an efficient instance.

Code-Based Dithered Quantization We adapt lattice-based dithered quantization to the binary field $\mathbb{F}_2^N$ by replacing Euclidean lattices with binary linear codes $\mathcal{C}$. Consider a quantization function $Q_{\mathcal{C}}(\cdot) : \mathbb{F}_2^N \rightarrow \mathcal{C}$. Together with dithering, the following theorem shows that the quantization error admits a uniform distribution over $\mathbb{F}_2^N/\mathcal{C}$ and is independent of the input.

Theorem 1 (Input Independence of Dithered Quantization Error). *Let $\mathcal{C} \subseteq \mathbb{F}_2^N$ be a binary linear code of dimension K with coset space $\mathbb{F}_2^N/\mathcal{C}$, $\mathbf{u} \sim \text{Unif}(\mathbb{F}_2^N/\mathcal{C})$ a uniformly random dither, and $\mathbf{y} \in \mathbb{F}_2^N$ any input vector. The quantization error*

$$\mathbf{e}_Q := Q_{\mathcal{C}}(\mathbf{y} \oplus \mathbf{u}) \oplus (\mathbf{y} \oplus \mathbf{u})$$

satisfies:

1. *Uniform distribution:* $\mathbf{e}_Q \sim \text{Unif}(\mathbb{F}_2^N/\mathcal{C})$;
2. *Statistical independence:* $\mathbf{e}_Q \perp \mathbf{x}$.

The notation “ $\perp$ ” denotes statistical independence.

Proof. To begin, we show that the coset uniformity is preserved. For any fixed $\mathbf{x}$, consider the shifted dither:

$$\mathbf{u}' := \mathbf{y} \oplus \mathbf{u}$$

The mapping $\phi_{\mathbf{x}} : \mathbb{F}_2^N/\mathcal{C} \rightarrow \mathbb{F}_2^N/\mathcal{C}$ defined by $\phi_{\mathbf{x}}(\mathbf{u}) = \mathbf{y} \oplus \mathbf{u}$ is a bijection on the coset space. By code quantization definition, the quantization error becomes:

$$\mathbf{e}_Q = Q_{\mathcal{C}}(\mathbf{u}') \oplus \mathbf{u}' = \text{Leader}(\mathbf{u}' \oplus \mathcal{C})$$

where $\text{Leader}(\mathcal{S})$ denotes the minimum Hamming weight representative of coset $\mathcal{S}$.

Since $\mathbf{u}'$ is uniform over 2^{N-K} cosets and each coset $\mathcal{S}$ has a unique leader, we have

$$\forall \mathbf{e} \in \mathbb{F}_2^N/\mathcal{C}, \quad \Pr[\mathbf{e}_Q = \mathbf{e}] = 2^{-(N-K)}$$

This probability mass function depends exclusively on $\mathcal{C}$'s structure, not on $\mathbf{x}$.

Lastly, for any $\mathbf{a} \in \mathbb{F}_2^N$ and $\mathbf{e} \in \mathbb{F}_2^N/\mathcal{C}$:

$$\Pr[\mathbf{y} = \mathbf{a} \wedge \mathbf{e}_Q = \mathbf{e}] = \Pr[\mathbf{y} = \mathbf{a}] \cdot \Pr[\mathbf{e}_Q = \mathbf{e}] = \Pr[\mathbf{y} = \mathbf{a}] \cdot 2^{-(N-K)}$$

The factorization demonstrates statistical independence $\mathbf{e}_Q \perp \mathbf{x}$. $\square$

The above theorem extends naturally to the case where $\mathbf{u} \sim \text{Unif}(\mathbb{F}_2^N)$ (full uniform dither). To see this: For any fixed input $\mathbf{y} \in \mathbb{F}_2^N$, the shifted dither $\mathbf{u}' := \mathbf{y} \oplus \mathbf{u}$ remains uniformly distributed over $\mathbb{F}_2^N$ by the translation invariance

of the uniform distribution. Hence its coset $\mathbf{u}' \oplus \mathcal{C}$ is uniformly distributed among all 2^{N-K} cosets. Consequently $\mathbf{e}_Q$ is uniformly distributed over the set of coset leaders of $\mathbb{F}_2^N/\mathcal{C}$ and $\mathbf{e}_Q \perp \mathbf{x}$. This generalizes Theorem 1 while preserving all essential properties.

Security via Rate-Distortion Optimality For LPQ, the input to the quantizer can be treated as a uniform source: Let $\mathbf{s} \in \mathbb{F}_2^n$ be fixed with $\mathbf{s} \neq \mathbf{0}$, and let $\mathbf{A} \in \mathbb{F}_2^{N \times n}$ have i.i.d. uniform entries. Then $\mathbf{b} = \mathbf{A}\mathbf{s} \in \mathbb{F}_2^N$ is uniformly distributed over $\mathbb{F}_2^N$. The quantizer's input in LPQ is $\mathbf{A}\mathbf{s} \oplus \mathbf{u}$ with $\mathbf{u} \sim \text{Unif}(\mathbb{F}_2^N/\mathcal{C})$. For uniform source, the rate-distortion optimal codes inherently produce Bernoulli-like quantization errors.

Theorem 2 (Distribution of the Quantization Error). *Let $Y^N = (Y_1, \dots, Y_N)$ be i.i.d. with $Y_i \sim \text{Bern}(\frac{1}{2})$, and let $\mathcal{C} \subseteq \mathbb{F}_2^N$ be a sequence of binary codes of rate*

$$R = \frac{1}{N} \log_2 |\mathcal{C}| \rightarrow R(\mu) = 1 - H(\mu),$$

as $N \rightarrow \infty$, where $H(\mu)$ is the binary entropy function and $\mu \in (0, \frac{1}{2})$. Assume $\mathcal{C}$ achieves the rate-distortion bound with quantizer $Q_{\mathcal{C}} : \mathbb{F}_2^N \rightarrow \mathcal{C}$ at an average distortion (normalized by N) μ under Hamming distortion as $N \rightarrow \infty$. Define the error vector

$$\mathbf{e}_Q = Y^N \oplus Q_{\mathcal{C}}(Y^N) \in \mathbb{F}_2^N.$$

Then

$$\Pr[\mathbf{e}_Q = e] = \mu^{\text{wt}(e)} (1 - \mu)^{N - \text{wt}(e)}, \quad \forall e \in \mathbb{F}_2^N,$$

i.e. $\mathbf{e}_Q \sim \text{Bern}(\mu)^N$ as $N \rightarrow \infty$.

Proof. For a Bernoulli($\frac{1}{2}$) source $Y^N \sim \text{Bern}(\frac{1}{2})^N$, the rate-distortion function is:

$$R(\mu) = 1 - H(\mu), \quad 0 \leq \mu \leq \frac{1}{2}. \quad (2)$$

By Shannon's rate-distortion theorem for the binary symmetric source under Hamming distortion, the optimal test channel ($W : \hat{Y} \rightarrow Y$) achieving $R(\mu) = 1 - H(\mu)$ at average distortion $\mathbb{E}[\hat{Y} \oplus Y] = \mu$ is the binary symmetric channel (BSC) with crossover probability μ [Cov99]. Specifically, we have

$$R(\mu) = \min_{p(\hat{Y}^N | Y^N) : \mathbb{E}[d(Y^N, \hat{Y}^N)] \leq \mu} \frac{1}{N} I(Y^N; \hat{Y}^N).$$

where $I(Y^N; \hat{Y}^N) = H(Y^N) - H(Y^N | \hat{Y}^N) = N - H(Y^N | \hat{Y}^N)$. Letting $H(Y^N | \hat{Y}^N) = N \cdot H(\mu)$, the rate distortion bound is achieved. Finally, $H(Y^N | \hat{Y}^N) = N \cdot H(\mu)$ holds iff $\hat{Y}^N = Y^N \oplus Z^N$ with $Z \sim \text{Bern}(\mu)$, $Z^N \perp Y^N$ (BSC structure).

The theorem states that the error vector $\mathbf{e}_Q = Y^N \oplus Q_{\mathcal{C}}(Y^N)$ satisfies:

$$\Pr[\mathbf{e}_Q = e] = \mu^{\text{wt}(e)} (1 - \mu)^{N - \text{wt}(e)}, \quad \forall e \in \mathbb{F}_2^N,$$

i.e., $\mathbf{e}_Q \sim \text{Bern}(\mu)^N$. This aligns with the $\text{BSC}(\mu)$ test channel

□

Polar Code Instantiation At the core of our framework lies the use of polar codes [Ari09] optimized for BSC with crossover probability μ . These codes achieve the theoretical rate-distortion bound $R(\mu) = 1 - H_2(\mu)$ through the channel polarization phenomenon, where $N = 2^r$ synthesized channels split into perfectly reliable or completely noisy subsets. The polarization process ensures that asymptotically, a fraction $1 - H_2(\mu)$ of channels become noiseless, enabling optimal compression while preserving distortion constraints.

Encoding and decoding are implemented via successive cancellation (SC), a recursive algorithm that exploits the Kronecker-based structure of polar codes to achieve $O(N \log N)$ complexity. This efficiency stems from the butterfly operations in the polarization transform, which decompose the code into smaller subcodes that are recursively processed. For distortion control, the quantization error $\mathbf{e}_Q$ satisfies $\mathbb{E}[\text{wt}(\mathbf{e}_Q)] = \mu N + o(N)$ with high probability, as polarized channels concentrate distortion near the target μ under SC decoding.

By integrating these properties, LPQ inherits cryptographic security through the computational indistinguishability of $\mathbf{e}_Q$ from Bernoulli noise (Theorem 2), matching the hardness assumptions of LPN. Simultaneously, polar codes enable practical efficiency: their near-linear complexity aligns with LPN-based cryptosystems while introducing no asymptotic overhead. Furthermore, syndrome compression of $\mathbf{e}_Q$ —exploiting the code’s parity-check structure—reduces ciphertext size, achieving a ciphertext rate $R \rightarrow 1$ as N grows.

1.3 Related Work

LPN in Cryptography The LPN problem has become a linchpin for constructing cryptographic primitives that balance efficiency with post-quantum security. Its appeal stems from three intrinsic properties: minimal computational overhead, simplicity of implementation, and resistance to known quantum attacks. In symmetric cryptography, most schemes rely on the constant-noise LPN variant, where the noise rate μ remains independent of the secret size n . Early work by Gilbert et al. [GRS08] established the LPN-C encryption scheme, reducing security directly to the LPN problem. Subsequent refinements by Applebaum et al. [App16] and Ben-Efraim et al. [BECO⁺21] demonstrated how LPN encryption enables secure garbling of circuits. Parallel efforts produced lightweight authentication protocols such as the HB family (HB [HB01], HB+ [JW05], HB++ [BCD06]), culminating in Kiltz et al.’s AUTH protocol [KPV⁺17], which achieves security against man-in-the-middle adversaries.

Public-key constructions, in contrast, necessitate low-noise regimes ($\mu \leq 1/\sqrt{n}$) to enable security reductions. Alekhnovich’s seminal work [Ale03] proposed the first public-key cryptosystem based on decisional LPN with constant noise. Damgård and Sunoo [DP12] later showed that transposed ring-LPN variants outperform their ring-LPN counterparts in practical parameter regimes. Döttling [Döt16] advanced this line by constructing the first standard-model

public-key encryption (PKE) with key-dependent message (KDM) security under low-noise LPN. Duc and Vaudenay’s HELEN [DV13] further operationalized these ideas, providing concrete parameter sets for real-world deployment.

Hardness and Attacks The cryptographic hardness of LPN is deeply intertwined with its significance in learning theory. Solving LPN would imply breakthroughs in learning 2-DNF formulas, juntas, and functions with sparse Fourier spectra [FGKP06], underscoring its foundational complexity. While Grover’s algorithm [Gro96] provides a quadratic speedup for brute-force attacks, no known quantum algorithm achieves an exponential improvement over the best classical methods for LPN [TV22].

Attacks on LPN bifurcate based on the noise regime. For constant noise $\mu = \Theta(1)$, the Blum-Kalai-Wasserman (BKW) algorithm [BKW03] dominates with $2^{O(n/\log n)}$ time, memory, and sample complexity. Over two decades, BKW has been optimized via advanced fast correlation attacks [FMI⁺06], covering codes [GJL14], and cascaded concrete perfect codes [ZJW16], with recent refinements reducing practical overheads [WS21]. In low-noise regimes ($\mu \leq 1/\sqrt{n}$), best-known attacks exploit structured guessing or correlation heuristics to achieve subexponential $2^{\tilde{O}(n^{1/2-\delta})}$ runtimes [BCGI18, BCG⁺20]. Theoretical underpinnings for LPN’s hardness have coalesced recently, with Brakerski et al. [BLVW19] establishing worst-case reductions for high-noise LPN and Yu et al. [YZ21] extending this to constant-noise settings via smoothing techniques.

Connection to LWE and LWQ LPN and the proposed variant LPQ represent binary-modulus specializations of the Learning With Errors (LWE) [Reg05] and Learning With Quantization (LWQ) [LLL24] problems. This relationship is nontrivial: while LWE/LWQ leverage large moduli and discrete Gaussian noise to enable lattice-based security proofs, LPN/LPQ restrict to modulus 2 and Bernoulli noise, necessitating parity-centric hardness assumptions. In LWQ, provably secure quantization relies on multilevel polar lattices derived from nested polar codes, whereas LPQ adopts a single-layer polar code approach. Both paradigms inject noise via lossy compression (LWQ/LPQ) or additive perturbations (LWE/LPN), but their security foundations diverge sharply—LWE inherits reductions from worst-case lattice problems, while LPN hardness remains grounded in combinatorial learning assumptions.

1.4 Paper Organization

The remainder of this paper is structured as follows. Section 2 presents basics of code-based quantization and polar codes. Section 3 introduces our proposed LPQ construction, including a formal security reduction to the LPN problem. Section 4 details a private-key encryption scheme achieving optimal rate efficiency. Section 5 compares LPQ and LPN in terms of rate efficiency, highlighting LPQ’s superior performance in high-noise settings. The last section concludes this paper.

2 Preliminaries

2.1 Code-Based Quantization

Let $\mathcal{C} \subset \mathbb{F}_2^N$ be a binary linear code of dimension K and length N . The space $\mathbb{F}_2^N$ is partitioned into cosets of the form $\mathbf{u} \oplus \mathcal{C} = \{\mathbf{u} \oplus \mathbf{c} : \mathbf{c} \in \mathcal{C}\}$, where $\mathbf{u} \in \mathbb{F}_2^N / \mathcal{C}$ is a representative of the coset and $\oplus$ denotes bitwise XOR. There are 2^{N-K} such cosets, each containing 2^K codewords.

A finite set $\mathcal{R}(\mathcal{C}) \subset \mathbb{F}_2^N$ is called a fundamental domain of $\mathcal{C}$ if $\cup_{\mathbf{c} \in \mathcal{C}} (\mathcal{R}(\mathcal{C}) \oplus \mathbf{c}) = \mathbb{F}_2^N$ and if $(\mathcal{R}(\mathcal{C}) \oplus \mathbf{c}) \cap (\mathcal{R}(\mathcal{C}) \oplus \mathbf{c}') = \emptyset$ for any $\mathbf{c} \neq \mathbf{c}'$ in $\mathcal{C}$. The size of a fundamental domain is 2^{N-K} . For input $\mathbf{y} \in \mathbb{F}_2^N$, a code-based quantizer $\mathbf{Q}_{\mathcal{C}}(\cdot)$ associated with $\mathcal{R}(\mathcal{C})$ outputs

$$\mathbf{Q}_{\mathcal{C}}(\mathbf{y}) = \mathbf{c}, \quad \forall \mathbf{y} \in \mathcal{R}(\mathcal{C}) \oplus \mathbf{c}. \quad (3)$$

Note that the quantization rule is not unique: a different $\mathcal{R}(\mathcal{C})$ is associated with a different quantizer. In particular, the nearest neighbor quantizer maps each vector $\mathbf{y}$ to the closest codeword in $\mathcal{C}$ under the Hamming distance:

$$\mathbf{Q}_{\mathcal{C}}(\mathbf{y}) = \arg \min_{\mathbf{c} \in \mathcal{C}} d_H(\mathbf{y}, \mathbf{c}),$$

where $d_H(\cdot, \cdot)$ denotes the Hamming distance. In practice, however, a more general quantizer (3) is often used due to efficiency reasons. For instance, quantization can be performed efficiently for modern codes such as LDPC [RU08] or polar codes [Ari09].

The main drawback of deterministic quantization given above is that the quantization error $(\mathbf{y} \oplus \mathbf{Q}_{\mathcal{C}}(\mathbf{y}))$ depends on the statistics of the input $\mathbf{y}$. To randomize the quantization process, we introduce a public random vector $\mathbf{u} \sim \text{Unif}(\mathbb{F}_2^N / \mathcal{C})$, i.e., $\mathbf{u}$ is chosen uniformly at random from a complete set of coset representatives of $\mathbb{F}_2^N$ modulo $\mathcal{C}$. Formally, define the dithered quantizer $\mathbf{Q}_{\mathcal{C}, \mathbf{u}} : \mathbb{F}_2^N \rightarrow \mathbf{u} \oplus \mathcal{C}$ by

$$\mathbf{Q}_{\mathcal{C}, \mathbf{u}}(\mathbf{y}) = \mathbf{Q}_{\mathcal{C}}(\mathbf{y} - \mathbf{u}) \oplus \mathbf{u}. \quad (4)$$

The advantage of this approach is that the quantization error $\mathbf{e}_Q = \mathbf{y} \oplus \mathbf{Q}_{\mathcal{C}, \mathbf{u}}(\mathbf{y})$ satisfies $\mathbf{e}_Q \sim \text{Unif}(\mathbb{F}_2^N / \mathcal{C})$, and it holds that $\mathbf{e}_Q \perp \mathbf{y}$ as shown in Theorem 1. If $\mathcal{C}$ is a good covering code, the average Hamming distortion $D = \mathbb{E}[\|\mathbf{e}_Q\|_1] / N$ can be made small. As the dither $\mathbf{u}$ is public, the quantization rate is the same as the rate of the code $\mathcal{C}$: $R = K/N$.

2.2 Polar Codes

Polar codes achieve capacity via channel polarization [Ari09]. For $N = 2^r$, the transform $\mathbf{G}_N = (\mathbf{B}_N \mathbf{F}^{\otimes r})^\top$, where $\mathbf{F} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $\mathbf{B}_N$ is the bit-reversal permutation matrix, synthesizes N channels $\{W_N^{(i)}\}$ with polarized reliabilities char-

acterized by their Bhattacharyya parameters:

$$Z(W_N^{(i)}) := \sum_{y \in \mathcal{Y}} \sqrt{W_N^{(i)}(y|0)W_N^{(i)}(y|1)}$$

where $W_N^{(i)}(y|x)$ is the channel transition probability. These parameters satisfy:

- $Z(W_N^{(i)}) \approx 0$: High-reliability channel (carries information)
- $Z(W_N^{(i)}) \approx 1$: Low-reliability channel (frozen to 0)

The effective $N \times K$ generator matrix $\mathbf{G}$ formed by selecting columns of $\mathbf{G}_N$ corresponding to channels with smallest $Z(W_N^{(i)})$. The code rate $R = K/N \approx 1 - H(\mu)$ for BSC(μ), where K equals the number of channels with high reliabilities ($Z(W_N^{(i)}) < 2^{-N^\beta}$) for $\beta < 1/2$.

Quantization based on polar codes: Here, we briefly describe the successive cancellation (SC) decoding algorithm. Assume that the polar code is constructed according to the binary-input test channel $W : X \rightarrow Y$ (the forward channel is $V : Y \rightarrow X$). Let the source sequence be denoted by $y^{[N]}$ and its reconstruction by $x^{[N]}$. The encoded sequence is given by $x^{[N]} = \mathbf{G}_N u^{[N]}$, with the frozen bits fixed as $u^{\mathcal{F}}$. The decoder's task is to estimate $\hat{u}^{[N]}$ based on $y^{[N]}$.

The i -th polarized subchannel is defined by the following probability:

$$W_N^{(i)}(y^{[N]}, u^{1:i-1}|u^i) \triangleq \sum_{u^{i+1:N}} \frac{1}{2^{N-1}} W_N(y^{[N]}|u^{[N]}),$$

where $W_N(y^{[N]}|u^{[N]}) = P_{Y^{[N]}|X^{[N]}}(y^{[N]}|\mathbf{G}_N u^{[N]})$.

The SC decoder utilizes the Log-Likelihood Ratio (LLR) defined as

$$L_N^{(i)}(y^{[N]}, \hat{u}_1^{i-1}) \triangleq \log_e \frac{W_N^{(i)}(y^{[N]}, \hat{u}_1^{i-1}|0)}{W_N^{(i)}(y^{[N]}, \hat{u}_1^{i-1}|1)}.$$

Let $\mathbf{f}(a, b) \triangleq \log_e \left(\frac{e^a + e^b}{e^a + e^b} \right)$ and $\mathbf{g}(a, b, \lambda) \triangleq (-1)^\lambda a + b$. The LLR can be computed recursively as:

$$\begin{aligned} & L_N^{(2i-1)}(y^{[N]}, \hat{u}_1^{1:2i-2}) \\ &= \mathbf{f} \left(L_{N/2}^{(i)}(y^{[N/2]}, \hat{u}_o^{1:2i-2} \oplus \hat{u}_e^{1:2i-2}), L_{N/2}^{(i)}(y^{N/2+1:N}, \hat{u}_e^{1:2i-2}) \right), \end{aligned} \quad (5)$$

and

$$\begin{aligned} & L_N^{(2i)}(y^{[N]}, \hat{u}_1^{1:2i-1}) \\ &= \mathbf{g} \left(L_{N/2}^{(i)}(y^{[N/2]}, \hat{u}_o^{1:2i-2} \oplus \hat{u}_e^{1:2i-2}), L_{N/2}^{(i)}(y^{N/2+1:N}, \hat{u}_e^{1:2i-2}), \hat{u}_1^{1:2i-1} \right), \end{aligned} \quad (6)$$

where $\hat{u}_o^{1:2i-2}$ and $\hat{u}_e^{1:2i-2}$ represent the odd and even indexed subvectors of $\hat{u}^{1:2i}$, respectively.

The Maximum A Posteriori (MAP) decoding rule for the i -th bit is:

$$\hat{u}^i = \begin{cases} u^i, & i \in \mathcal{F}, \\ 0, & i \in \mathcal{I}, L_N^{(i)}(y^{[N]}, \hat{u}^{1:i-1}) \geq 0, \\ 1, & i \in \mathcal{I}, L_N^{(i)}(y^{[N]}, \hat{u}^{1:i-1}) < 0. \end{cases} \quad (7)$$

Note that the SC decision rule in (7) is deterministic. However, for more convenient analysis, it can be modified to a random rounding version:

$$\hat{u}^i = \begin{cases} u^i, & i \in \mathcal{F}, \\ 0, & i \in \mathcal{I}, \text{ with probability } \frac{\exp(L_N^{(i)})}{1+\exp(L_N^{(i)})}, \\ 1, & i \in \mathcal{I}, \text{ with probability } \frac{1}{1+\exp(L_N^{(i)})}. \end{cases} \quad (8)$$

The performance difference between the deterministic and random rounding decision rules is negligible, as observed in [Kor09]. Frozen bits can be programmed with dither values to implement code-based dithered quantization. This approach preserves the polarization properties while introducing input-independent noise.

SC decoding has quasi-linear time complexity $O(N \log N)$.

2.3 LPN and Private-Key Encryption

Definition 1 (LPN: Decisional and Search Variants). Let $N, n \in \mathbb{N}$ and $\mu \in (0, 1/2)$. The LPN problem has two main formulations:

Decisional LPN ($LPN_{N,n,\mu}$): Given $(\mathbf{A}, \mathbf{b})$, determine whether it is drawn from the LPN distribution or the uniform distribution:

$$\mathcal{D}_{\text{LPN}} : \left\{ (\mathbf{A}, \mathbf{b}) : \mathbf{A} \xleftarrow{\$} \mathbb{F}_2^{N \times n}, \mathbf{s} \xleftarrow{\$} \{0, 1\}^n, \mathbf{e} \sim \text{Bern}(\mu)^N, \mathbf{b} = \mathbf{A}\mathbf{s} \oplus \mathbf{e} \right\}$$

$$\mathcal{D}_{\text{U}} : \left\{ (\mathbf{A}, \mathbf{b}) : \mathbf{A} \xleftarrow{\$} \mathbb{F}_2^{N \times n}, \mathbf{b} \xleftarrow{\$} \mathbb{F}_2^N \right\}.$$

Search LPN: Given $(\mathbf{A}, \mathbf{b})$ from $\mathcal{D}_{\text{LPN}}$, recover the secret $\mathbf{s} \in \{0, 1\}^n$.

The search and decisional variants of LPN are polynomially equivalent [KSS10]. The inherent difficulty in solving either variant underpins the security of cryptographic protocols that utilize LPN. With these definitions in hand, we can now proceed to explore how LPN forms the basis of the private-key encryption schemes discussed in this paper.

Definition 2 (Standard LPN Encryption, adapted from [ACPS09]). Let $n, N = \text{poly}(\lambda)$ and $n < N$, where n denotes the secret length. Let $\mu \in (0, 1/2)$ be the noise rate and $\text{Bern}(\mu)^N$ the corresponding Bernoulli distribution over N bits. Moreover, let $\mathbf{G}_E \in \mathbb{F}_2^{N \times K}$ be a generator matrix of a binary error correction code $\mathcal{C}_E$. The standard LPN-based symmetric encryption scheme consists of the following algorithms:

- $\text{KeyGen}_\lambda(1^\lambda)$: On input the security parameter λ , sample a secret key $\mathbf{s} \xleftarrow{\$} \mathbb{F}_2^n$.
- $\text{Enc}_\lambda(\mathbf{m}, \mathbf{s})$: Given a message $\mathbf{m} \in \mathbb{F}_2^K$ (i.e., the plaintext pt) and a secret key $\mathbf{s} \in \mathbb{F}_2^n$, sample a matrix $\mathbf{A} \xleftarrow{\$} \mathbb{F}_2^{N \times n}$ and noise $\mathbf{e} \sim \text{Bern}(\mu)^N$, and output the ciphertext

$$ct = (\mathbf{A}, \mathbf{b} = \mathbf{A}\mathbf{s} \oplus \mathbf{e} \oplus \mathbf{G}_E \mathbf{m})$$

- $\text{Dec}_\lambda(ct, \mathbf{s})$: Given a ciphertext $ct = (\mathbf{A}, \mathbf{b})$ and the secret key $\mathbf{s}$, compute $\mathbf{b} \oplus \mathbf{A}\mathbf{s}$ and then apply the decoding algorithm for $\mathcal{C}_E$ to recover $\mathbf{m}$.

An extension of the above encryption scheme, where the message $\mathbf{m}$ is generalized from a vector in $\mathbb{F}_2^K$ to a matrix in $\mathbb{F}_2^{K \times \ell}$, was proven to be semantically secure. Moreover, it offers robustness against two advanced forms of cryptanalytic attacks: key-dependent message (KDM) attacks and related-key attacks (RKA), specifically for linear functions [ACPS09]. In KDM attacks, an adversary is allowed to obtain encryptions of functions of the secret key itself (e.g., $\text{Enc}_{\text{sk}}(f(\text{sk}))$), while RKA security considers scenarios where the adversary can query encryptions under keys related to the original secret key (e.g., $\text{sk}' = f(\text{sk})$ for some known f). The scheme was later shown to be secure under the stronger notion of *linear related-key key-dependent message* (LIN RK-KDM) security [App16], which simultaneously allows key relations and message dependencies, provided both are linear functions. This notion of security is especially important in cryptographic protocols like secure multiparty computation and program obfuscation, where adversaries may exploit key relations and key-dependent ciphertexts.

Theorem 3 ([App16]). *Assuming the hardness of $\text{DLPN}_{N,n,\lambda}$, the standard LPN encryption scheme is LIN-RK-KDM secure.*

Define the information rate of an encryption scheme as

$$R = \frac{\text{plaintext size } |\text{pt}|}{\text{ciphertext size } |\text{ct}|}.$$

In our constructions, an eXtendable Output Function (XOF) $\text{XOF} : \{0, 1\}^* \rightarrow \mathbb{F}_2^{N \times n}$ is used to generate the matrix $\mathbf{A}$ via

$$\mathbf{A} \leftarrow \text{XOF}(\text{seed}),$$

where the seed is a fixed, small constant. As a result, the cost of storing $\mathbf{A}$ is effectively constant, and we do not count its size when determining the rate.

For LPN-based symmetric encryption schemes, the achievable information rate is fundamentally limited by the noise introduced in the encryption process. In particular, the rate is upper-bounded by the BSC capacity. Consequently, the maximum possible information rate for LPN-based encryption is

$$R \leq C = 1 - H_2(\mu).$$

3 The Proposed LPQ Family

We formally define the LPQ problem. Unlike LPN where noise terms must remain secret, the noise of LPQ originates from quantization with a *public dither* $\mathbf{u}$, enabling efficient correction at the receiver while preserving security. We also introduce a variant LPQ' to serve the security proof of LPQ.

Definition 3 (LPQ' and LPQ). *Let $N, n \in \mathbb{N}$ and let $\mathcal{C}$ be an (N, K) quantization code. The problem has two formulations, depending on whether the dither $\mathbf{u}$ is presented explicitly:*

Decisional LPQ': Given $(\mathbf{A}, \mathbf{b})$, distinguish between:

$$\mathcal{D}_{\text{LPQ}'} : \left\{ (\mathbf{A}, \mathbf{b}) \mid \mathbf{A} \xleftarrow{\$} \mathbb{F}_2^{N \times n}, \mathbf{s} \xleftarrow{\$} \mathbb{F}_2^n, \mathbf{u} \xleftarrow{\$} \mathbb{F}_2^N / \mathcal{C}, \mathbf{b} = Q_{\mathcal{C}}(\mathbf{A}\mathbf{s} \oplus \mathbf{u}) \oplus \mathbf{u} \right\},$$

$$\mathcal{D}_{\text{U}'} : \left\{ (\mathbf{A}, \mathbf{b}) \mid \mathbf{A} \xleftarrow{\$} \mathbb{F}_2^{N \times n}, \mathbf{b} \xleftarrow{\$} \mathbb{F}_2^N \right\}.$$

Decisional LPQ: Given $(\mathbf{A}, \mathbf{b}, \mathbf{u})$, distinguish between:

$$\mathcal{D}_{\text{LPQ}} : \left\{ (\mathbf{A}, \mathbf{b}, \mathbf{u}) \mid \mathbf{A} \xleftarrow{\$} \mathbb{F}_2^{N \times n}, \mathbf{s} \xleftarrow{\$} \mathbb{F}_2^n, \mathbf{u} \xleftarrow{\$} \mathbb{F}_2^N / \mathcal{C}, \mathbf{b} = Q_{\mathcal{C}}(\mathbf{A}\mathbf{s} \oplus \mathbf{u}) \right\},$$

$$\mathcal{D}_{\text{U}} : \left\{ (\mathbf{A}, \mathbf{b}, \mathbf{u}) \mid \mathbf{A} \xleftarrow{\$} \mathbb{F}_2^{N \times n}, \mathbf{b} \xleftarrow{\$} \mathcal{C}, \mathbf{u} \xleftarrow{\$} \mathbb{F}_2^N / \mathcal{C} \right\}.$$

Search LPQ'/LPQ: Given samples from $\mathcal{D}_{\text{LPQ}'}$ (or $\mathcal{D}_{\text{LPQ}}$), recover the secret $\mathbf{s} \in \mathbb{F}_2^n$.

Lemma 1 (Equivalence of LPQ' and LPQ). *Assuming the quantization algorithm $Q_{\mathcal{C}}$ runs in polynomial time. LPQ' and LPQ are polynomial-time equivalent in both the decisional and search variants.*

Proof. Define the maps

$$g: (\mathbf{A}, \mathbf{b}, \mathbf{u}) \mapsto (\mathbf{A}, \mathbf{b} \oplus \mathbf{u}), \quad h: (\mathbf{A}, \mathbf{b}) \mapsto (\mathbf{A}, Q_{\mathcal{C}}(\mathbf{b}), \mathbf{b} \oplus Q_{\mathcal{C}}(\mathbf{b})).$$

Then

$$g(\mathcal{D}_{\text{LPQ}}) = \mathcal{D}_{\text{LPQ}'}, \quad g(\mathcal{D}_{\text{U}'}) = \mathcal{D}_{\text{U}}, \quad h(\mathcal{D}_{\text{LPQ}'}) = \mathcal{D}_{\text{LPQ}}, \quad h(\mathcal{D}_{\text{U}}) = \mathcal{D}_{\text{U}'}.$$

Both g and h run in polynomial time. Therefore any polynomial-time distinguisher or solver for one distribution can be converted into one for the other. $\square$

Advantage: Storage Efficiency By generating $\mathbf{A}$ (in LPN) and $(\mathbf{A}, \mathbf{u})$ (in LPQ) deterministically from a short random seed using an extendable output function (XOF), the only remaining randomness that must be stored is mixture vector $\mathbf{b}$. In standard LPN, this requires storing the full vector $\mathbf{b} = \mathbf{A}\mathbf{s} \oplus \mathbf{e} \in \mathbb{F}_2^N$, resulting in N bits of storage. In contrast, we have $\mathbf{b} = Q_{\mathcal{C}}(\mathbf{A}\mathbf{s} \oplus \mathbf{u})$ for LPQ, which consumes only K bits for storage.

Quantization Error Characterization. For polar codes, the quantization error $\mathbf{e}_Q = Q_C(\mathbf{A}\mathbf{s} \oplus \mathbf{u}) \oplus \mathbf{A}\mathbf{s} \oplus \mathbf{u}$ mimics i.i.d. Bernoulli noise while the computational complexity of the quantizer remains polynomial.

Theorem 4 (Statistical Indistinguishability of $\mathbf{e}_Q$). *Let $N = 2^r$ and C be a polar code for quantization with target distortion μ . For any $\beta < \frac{1}{2}$, there exists $c > 0$ such that for large N :*

$$d_{TV}(\mathbf{e}_Q, \text{Bern}(\mu)^N) \leq c \cdot 2^{-N^\beta},$$

where d_{TV} is the total variation distance, defined as: $d_{TV}(P, Q) := \frac{1}{2} \sum_x |P(x) - Q(x)|$.

Proof. The polar transform $\mathbf{G}_N = (\mathbf{B}_N \mathbf{F}^{\otimes r})^\top$ creates N synthetic channels. By Arikan's polarization theorem, for $\beta' < \frac{1}{2}$, the non-polarized reliabilities $Z(W_N^{(i)})$ satisfies:

$$\lim_{N \rightarrow \infty} \frac{|\{i : Z(W_N^{(i)}) \in (2^{-N^{\beta'}}, 1 - 2^{-N^{\beta'}})\}|}{N} = 0.$$

The frozen set is then defined from the channels with low reliabilities:

$$\mathcal{F}_N \triangleq \{i : Z(W_N^{(i)}) > 1 - 2^{-N^{\beta'}}\},$$

and the information set $\mathcal{I}_N$ is the complement set of $\mathcal{F}_N$.

Frozen bits (indexed by $U^{\mathcal{F}_N}$) are set to public dither values since the mutual information between them and the source vector is negligible, i.e., the synthetic polarized channels between the frozen bits and the source vector are almost completely noisy. For information bits ($U^{\mathcal{I}_N}$), their correlation with the source vector is significant and hence they are determined by the SC random rounding rule (8) according to the source realization. Therefore, the distribution distance between $\mathbf{e}_Q$ and $\text{Bern}(\mu)^N$ is caused by replacing $U^{\mathcal{F}_N}$ with public dither bits, which contributes to the total variance distance as follows [KU10]:

$$d_{TV}(\mathbf{e}_Q, \text{Bern}(\mu)^N) \leq \sum_{i \in \mathcal{F}_N} \sqrt{2 \log_e 2 \left(1 - Z(W_N^{(i)})^2\right)} \leq c \cdot 2^{-N^\beta},$$

for any $0 < \beta < \beta'$ and a sufficiently large N . $\square$

Theorem 5 (LPQ' $\approx_s$ LPN). *For polar-code based LPQ' samples $(\mathbf{A}', \mathbf{b}')$ and LPN samples $(\mathbf{A}, \mathbf{b})$, we have*

$$d_{TV}((\mathbf{A}', \mathbf{b}'), (\mathbf{A}, \mathbf{b})) \leq c \cdot 2^{-N^\beta}.$$

Proof. Let

$$P = \mathcal{D}(\mathbf{A}', \mathbf{s}', \mathbf{e}_Q), \quad Q = \mathcal{D}(\mathbf{A}, \mathbf{s}, \mathbf{e}),$$

where

$$\mathbf{A}, \mathbf{A}' \stackrel{\$}{\sim} \mathbb{F}_2^{N \times n}, \quad \mathbf{s}, \mathbf{s}' \stackrel{\$}{\sim} \mathbb{F}_2^n,$$

are independent of the noise vectors $\mathbf{e}_Q$ (the LPQ' quantization error) and $\mathbf{e} \sim \text{Bern}(\mu)^N$ (the LPN error). Define the deterministic map

$$f : \mathbb{F}_2^{N \times n} \times \mathbb{F}_2^n \times \mathbb{F}_2^N \longrightarrow \mathbb{F}_2^{N \times n} \times \mathbb{F}_2^N, \quad f(\mathbf{M}, \mathbf{x}, \mathbf{y}) = (\mathbf{M}, \mathbf{M}\mathbf{x} \oplus \mathbf{y}).$$

Then

$$(\mathbf{A}', \mathbf{b}') = f(\mathbf{A}', \mathbf{s}', \mathbf{e}_Q), \quad (\mathbf{A}, \mathbf{b}) = f(\mathbf{A}, \mathbf{s}, \mathbf{e}).$$

By the *data-processing inequality* for total variation distance,

$$d_{\text{TV}}(f(P), f(Q)) \leq d_{\text{TV}}(P, Q).$$

Moreover, since in both distributions the pair $(\mathbf{A}', \mathbf{s}')$ (resp. $(\mathbf{A}, \mathbf{s})$) is independent of the noise,

$$d_{\text{TV}}(P, Q) = d_{\text{TV}}(\mathbf{e}_Q, \mathbf{e}).$$

Applying Theorem 4 then gives

$$d_{\text{TV}}((\mathbf{A}', \mathbf{b}'), (\mathbf{A}, \mathbf{b})) \leq d_{\text{TV}}(\mathbf{e}_Q, \mathbf{e}) \leq c \cdot 2^{-N^\beta},$$

as required. $\square$

Theorem 6 (LPQ $\approx_c$ Uniform). *Assume that*

- *Decisional-LPN at error rate $\mu \in (0, \frac{1}{2})$ is hard (advantage $\leq \text{negl}_{\text{LPN}}(N)$).*
- *Polar codes $\{\mathcal{C}_N\}$ for $\text{BSC}(\mu)$ satisfy Theorem 4 (statistical distance $\leq \text{negl}_s(N)$).*

Then LPQ samples are computationally indistinguishable from uniform

$$\mathcal{D}_{\text{LPQ}} \approx_c \mathcal{D}_{\text{U}},$$

with advantage $\leq \text{negl}_{\text{LPN}}(N) + \text{negl}_s(N)$.

Proof. By Theorem 5,

$$\mathcal{D}_{\text{LPQ}'} \approx_s \text{LPN} \quad (\text{adv} \leq \text{negl}_s(N)),$$

and by decisional-LPN,

$$\text{LPN} \approx_c \mathcal{D}_{\text{U}'} \quad (\text{adv} \leq \text{negl}_{\text{LPN}}(N)).$$

Hence $\mathcal{D}_{\text{LPQ}'} \approx_c \mathcal{D}_{\text{U}'}$ with total advantage $\leq \text{negl}_s(N) + \text{negl}_{\text{LPN}}(N)$. Together with Lemma 1, the theorem is proved. $\square$

4 The XOF–LPQ Encryption Scheme

In our scheme the code $\mathcal{C}$ serves the purpose of quantization, rather than error correction. We use an XOF to generate both $\mathbf{A}$ and the public dither $\mathbf{u}$ from a short seed. Concretely, let $(\mathbf{A}, \mathbf{u}) = \text{XOF}(\text{seed})$ and $\mathbf{G}$ be the generator matrix

of a quantization code $\mathcal{C}$. To encrypt a message $\mathbf{m}$ based on secret key $\mathbf{s}$, we compute

$$\mathbf{b} = Q_{\mathcal{C}}(\mathbf{A}\mathbf{s} \oplus \mathbf{u}) \oplus \mathbf{G}\mathbf{m}$$

where $\mathbf{s}$ is the secret key, and $Q_{\mathcal{C}}(\mathbf{A}\mathbf{s} \oplus \mathbf{u})$ serves as the mask for both encryption and decryption. The ciphertext of message $\mathbf{m}$ is

$$\text{Enc}(\mathbf{m}, \mathbf{s}) : (\text{seed}, \mathbf{b}).$$

Decryption is done by computing $\mathbf{b} \oplus Q_{\mathcal{C}}(\mathbf{A}\mathbf{s} \oplus \mathbf{u})$.

Definition 4 (XOF-LPQ Encryption). Let $n, N = \text{poly}(\lambda)$ and $n < N$, where n denotes the secret length. Let $\mathcal{C}$ be a quantization code of parameter (N, K) , and $\mathbf{G} \in \mathbb{F}_2^{N \times K}$ be its generator matrix. The LPQ symmetric encryption scheme consists of the following algorithms:

- $\text{KeyGen}_{\lambda}(1^{\lambda})$: On input the security parameter λ , sample a secret key $\mathbf{s} \xleftarrow{\$} \mathbb{F}_2^n$.
- $\text{Enc}_{\lambda}(\mathbf{m}, \mathbf{s})$: Given a message $\mathbf{m} \in \mathbb{F}_2^K$ (i.e., the plaintext pt) and a secret key $\mathbf{s} \in \mathbb{F}_2^n$, generate $(\mathbf{A}, \mathbf{u}) = \text{XOF}(\text{seed})$ from the XOF of random seed, where $\mathbf{A} \in \mathbb{F}_2^{N \times n}$, $\mathbf{u} \in \mathbb{F}_2^N / \mathcal{C}$, and output the ciphertext

$$ct = (\text{seed}, \mathbf{b} = Q_{\mathcal{C}}(\mathbf{A}\mathbf{s} \oplus \mathbf{u}) \oplus \mathbf{G}\mathbf{m}).$$

- $\text{Dec}_{\lambda}(ct, \mathbf{s})$: Given a ciphertext $ct = (\text{seed}, \mathbf{b})$ and the secret key $\mathbf{s}$, let $(\mathbf{A}, \mathbf{u}) = \text{XOF}(\text{seed})$, the decrypted message is

$$\hat{\mathbf{m}} = \tilde{\mathbf{G}} \cdot (\mathbf{b} \oplus Q_{\mathcal{C}}(\mathbf{A}\mathbf{s} \oplus \mathbf{u})), \quad (9)$$

where $\tilde{\mathbf{G}}$ is the left-inverse of $\mathbf{G}$ (i.e., $\tilde{\mathbf{G}} \cdot \mathbf{G} = \mathbf{I}_K$).

The proposed LPQ-based encryption scheme achieves rate-optimal ciphertext compression by unifying error correction and encryption through structured quantization: excluding the size of *seed*, its information rate is $R = 1$, which is larger than the rate of $1 - H(\mu)$ based on LPN.

4.1 Correctness

By using the same mask for encryption and decryption, the decryption side has

$$\mathbf{b} \oplus Q_{\mathcal{C}}(\mathbf{A}\mathbf{s} \oplus \mathbf{u}) = \mathbf{G}\mathbf{m}. \quad (10)$$

Thus inverting this equation can obtain $\mathbf{m}$ correctly.

4.2 Security Proof

We prove that the XOF-LPQ encryption scheme is indistinguishability under chosen plaintext attack (IND-CPA) secure under the assumption that the LPQ problem is hard.

Definition 5 (IND-CPA Security, [KL07]). Let $\mathcal{SE} = (\text{KeyGen}, \text{Enc}, \text{Dec})$ be a symmetric encryption scheme with message space $\mathcal{M}$. $\mathcal{SE}$ is IND-CPA secure if for all probabilistic polynomial-time (PPT) adversaries $\mathcal{A}$, the following advantage is negligible in the security parameter λ :

$$\text{Adv}_{\mathcal{SE}, \mathcal{A}}^{\text{IND-CPA}}(\lambda) = \left| \Pr \left[\text{Exp}_{\mathcal{SE}, \mathcal{A}}^{\text{IND-CPA}}(\lambda) = 1 \right] - \frac{1}{2} \right|,$$

where the IND-CPA experiment $\text{Exp}_{\mathcal{SE}, \mathcal{A}}^{\text{IND-CPA}}(\lambda)$ is defined as follows:

1. *Key Generation:* The challenger generates a secret key $\mathbf{s} \leftarrow \text{KeyGen}(1^\lambda)$.
2. *Query Phase:* $\mathcal{A}$ may adaptively query an encryption oracle $\text{Enc}(\mathbf{s}, \cdot)$ to encrypt arbitrary messages $m \in \mathcal{M}$.
3. *Challenge Phase:* $\mathcal{A}$ submits two messages $m_0, m_1 \in \mathcal{M}$ of equal length. The challenger flips a bit $\beta \xleftarrow{\$} \{0, 1\}$, computes $ct^* \leftarrow \text{Enc}(\mathbf{s}, m_\beta)$, and returns ct^* to $\mathcal{A}$.
4. *Guess Phase:* $\mathcal{A}$ outputs a guess $\beta' \in \{0, 1\}$. The experiment outputs 1 if $\beta' = \beta$, and 0 otherwise.

Theorem 7. Let λ be the security parameter, and let $N, n = \text{poly}(\lambda)$. Suppose the LPQ problem is hard. Then, the XOF-LPQ encryption scheme is IND-CPA secure.

Proof. We formalize the reductionist security argument. Assume there exists a PPT adversary $\mathcal{A}$ that wins the IND-CPA game for the XOF-LPQ scheme with non-negligible advantage $\epsilon(\lambda)$. The proof is to construct a PPT distinguisher $\mathcal{S}$ that solves the LPQ problem with non-negligible advantage, contradicting the LPQ hardness assumption.

IND-CPA Game Simulation:

1. The LPQ challenger samples $\mathbf{A} \xleftarrow{\$} \mathbb{F}_2^{N \times n}$, $\mathbf{s} \xleftarrow{\$} \mathbb{F}_2^n$, $\mathbf{u} \xleftarrow{\$} \mathbb{F}_2^N / \mathcal{C}$, and flips a bit $\gamma \xleftarrow{\$} \{0, 1\}$. It returns:

$$(\mathbf{A}, \mathbf{b}', \mathbf{u}') = \begin{cases} (\mathbf{A}, Q_{\mathcal{C}}(\mathbf{A}\mathbf{s} \oplus \mathbf{u}), \mathbf{u}) & \text{if } \gamma = 0, \\ (\mathbf{A}, \mathbf{b}' \xleftarrow{\$} \mathcal{C}, \mathbf{u}' \xleftarrow{\$} \mathbb{F}_2^N / \mathcal{C}) & \text{if } \gamma = 1. \end{cases}$$

2. Distinguisher $\mathcal{S}$ receives $(\mathbf{A}, \mathbf{b}', \mathbf{u}')$ and simulates the IND-CPA game for $\mathcal{A}$:
When $\mathcal{A}$ submits messages $\mathbf{m}_0, \mathbf{m}_1$, $\mathcal{S}$ chooses $\beta \xleftarrow{\$} \{0, 1\}$ and computes:

$$\mathbf{b} = \mathbf{b}' \oplus \mathbf{G}\mathbf{m}_\beta,$$

where $\mathbf{G}$ generates a subcode of $\mathcal{C}$. $\mathcal{S}$ freshly samples seed seed^* , and programs the random oracle O as $O(\text{seed}^*) = (\mathbf{A}, \mathbf{u})$.

3. $\mathcal{A}$ outputs a guess β' , and $\mathcal{S}$ forwards $\gamma' = 1$ if $\beta' \neq \beta$, else $\gamma' = 0$.

The analysis of simulation is given as follows. Regarding Case 1 ($\gamma = 0$): $\mathcal{A}$'s view is identical to the real IND-CPA game. Regarding Case 2 ($\gamma = 1$): $\mathbf{b}'$ is

uniform over $\mathbb{F}_2^N$. Then $\mathbf{b} = \mathbf{b}' \oplus \mathbf{G}\mathbf{m}_\beta$ is also uniform, independent of β . Thus, $\mathcal{A}$'s advantage is 0.

In summary, $\mathcal{S}$ succeeds if $\gamma' = \gamma$. When $\gamma = 0$, $\mathcal{A}$ guesses β' with advantage $\epsilon(\lambda)$. When $\gamma = 1$, $\mathcal{A}$'s guess is random. Thus:

$$\Pr[\gamma' = \gamma] = \frac{1}{2} \cdot \left(\frac{1}{2} + \epsilon(\lambda) \right) + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2} + \frac{\epsilon(\lambda)}{2}.$$

$\mathcal{S}$'s advantage is $\epsilon(\lambda)/2$, which is non-negligible. This contradicts the LPQ hardness assumption, proving the theorem. $\square$

To protect against powerful adversaries capable of submitting ciphertexts to a decryption oracle, we require IND-CCA (indistinguishability under chosen ciphertext attack) security. Two standard approaches can be adopted to achieve IND-CCA security: (1) applying the Fujisaki-Okamoto (FO) transform [FO13, HHK17], which builds a non-malleable hybrid encryption scheme from a CPA-secure base scheme in the random oracle model (ROM); and (2) appending a message authentication code (MAC) [BN00] to the ciphertext to prevent tampering, which also yields IND-CCA security under standard assumptions. A separate proof of non-malleability is not required, since IND-CCA already ensures it.

5 Concrete Parameters

Recent analyses, such as those in [EKM17], suggest that the secret length n should be chosen to satisfy the following security condition:

$$n \geq \frac{\lambda}{\log_2 \left(\frac{1}{1-\mu} \right)},$$

where λ represents the symmetric-key security parameter, typically set to 128 or 256 bits, and μ is the noise rate in the LPN/LPQ setting.

In addition to the secret length n , it is crucial to evaluate the information rate of the corresponding cryptosystem. For LPN, assuming a capacity-achieving error-correcting code, the information rate (i.e., the ratio of the message length to the ciphertext length) is $R_{\text{LPN}} \approx 1 - H_2(\mu)$. For LPQ, the rate is $R_{\text{LPQ}} = 1$ as $\mathbf{G}$ generates the code $\mathcal{C}$.

Table 1 compares the security and efficiency trade-offs between LPN and LPQ schemes by analyzing two key parameters: (1) the minimum secret length n required to achieve λ -bit security against known attacks, and (2) the effective information rate. The results reveal a fundamental limitation of LPN-based cryptography: as the noise rate μ increases, the scheme suffers an exponential growth in required secret length (e.g., n escalates from 128 to 2,844 bits when μ decreases from 2^{-1} to 2^{-5} for $\lambda = 128$) while simultaneously experiencing severe rate degradation due to error correction overhead (dropping to $R = 0$

at $\mu = 0.5$). LPQ schemes completely circumvent this trade-off by eliminating the need for error correction, maintaining a perfect rate ($R = 1$) across all noise levels. The performance gap between LPQ and LPN widens dramatically in high-noise regimes ($\mu \geq 0.25$), underscoring LPQ’s superiority.

Noise μ	$-\log_2(1 - \mu)$	n for $\lambda = 128$	n for $\lambda = 256$	LPN Rate $1 - H_2(\mu)$	LPQ Rate
$2^{-1} = 0.5$	1.000	128	256	0.000	1
$2^{-2} = 0.25$	0.415	309	617	0.189	1
$2^{-3} = 0.125$	0.193	664	1328	0.456	1
$2^{-4} = 0.0625$	0.093	1376	2752	0.685	1
$2^{-5} = 0.03125$	0.045	2844	5688	0.846	1

Table 1. Comparison of LPN and LPQ parameters showing required secret length n and information rates.

6 Conclusions

In this paper, we introduced the LPQ problem as a new paradigm that unifies security with ciphertext compression. It is the analog of (lattice based) LWQ in binary fields. By leveraging quantization techniques and public dithers, LPQ not only retains the strong security guarantees inherited from LPN but also overcomes its inherent inefficiency in ciphertext size. Our construction achieves rate-1 encryption when combined with an XOF. While we focused on symmetric encryption, the LPQ framework can be extended to other cryptographic primitives, such as authentication protocols or key exchange mechanisms.

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